Optimum Machining Conditions for Mirror Polishing of Aluminum Alloy Using a Robot*

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Using robots in precision machining fields, such as grinding and polishing, is becoming more prominent in recent years. This paper presents original research results of aluminum alloy surface polishing using a rubber polishing disk in a robotic polishing system. A rotary hand-polishing system is held by the actuator of a six-axis articulated robot to perform the polishing task. The workpiece is an aluminum alloy block fixed on a piezo-electric type dynamometer. For finding the optimal tool angle to obtain the best polished surface, some polishing experiments were carried out using various tool angles. The effects of the number of polishing strokes on the surface roughness and depth of machining have been also estimated. Finally, calculated results and experimentally obtained results of a multi-stage polishing were compared. The obtained experimental results are useful for finding the optimum machining conditions for making an autonomous robotic polishing system.

Key Words: Robot, Polishing, Surface Roughness, Rubber Polishing Disk, Aluminum Alloy, Multi-Stage Polishing

1. Introduction

Presently, industrial robots are used in many areas of production industries. However, in fields of precision machining, such as mirror finishing, there are many difficulties to overcome for utilizing robotic automation. As a result, these kinds of tasks are often performed by skilled workers using hand-finishing systems with mounted grinding wheels or elastic polishing disks.

Aluminum alloy mirrors have become vital for many industries[1]. One of its usages is in the field of optics and astronomy. Some mirrors used in telescopes are composed of many aspheric aluminum mirror segments.

Another use of mirror-polished aluminum is found in the field of precision mould manufacturing[2]. The light weight, good thermal conductivity, and high shaping quality are some of the characteristics of aluminum. Therefore, it is used as a proper material for short production-cycle, high quality and low cost moulds. However, it is difficult to grind and polish the surfaces of aluminum alloy moulds because they are easily scratched.

Under these circumstances, this study intends to obtain the optimum machining conditions for mirror polishing of aluminum alloys in robotic polishing using a rubber polishing disk.

2. Robotic Polishing Method and Conditions

Figure 1 illustrates the experimental facilities used in this research work. The robot used was an articulated robot with six degrees of freedom with maximum accuracy of ± 0.01 mm in longitudinal movement and $\pm 0.01°$ in rotational one. The robot end-effector holds a hand-finishing head. The tool that is shown in Fig. 2 is a kind of polishing disk. As seen in this figure, there are inflected grooves on the disk surface, which has an inclination of $1° \pm 0.4°$ as shown by parameter β. The workpiece is an aluminum alloy block that is fixed on a piezo-electric type

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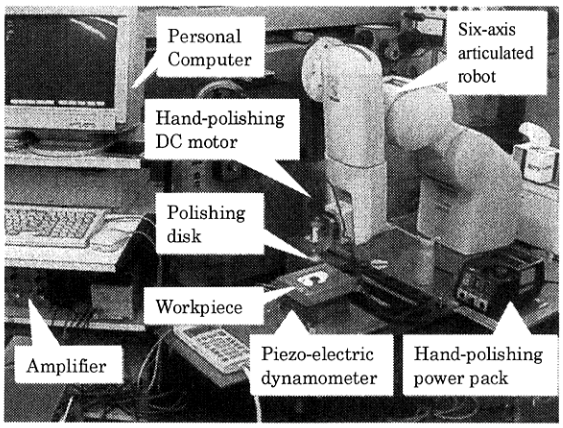
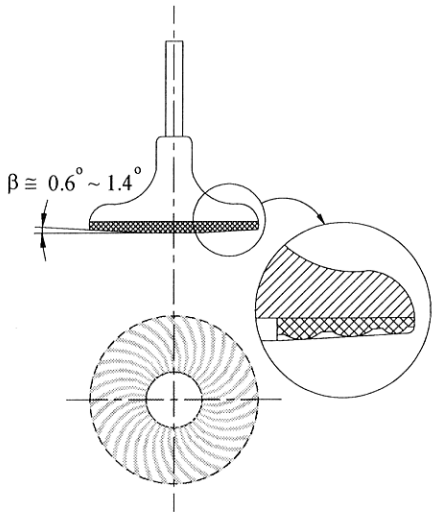
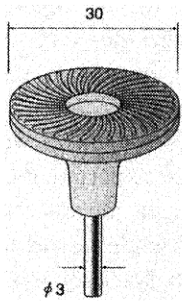


Fig. 1 Robotic polishing apparatus



(a) Detailed view



(b) General view

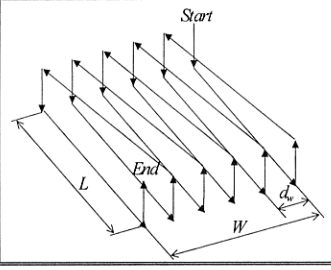
Fig. 2 The polishing disk

dynamometer. Three components of polishing force during the process are sent to a PC through an amplifier and A/D converter. The experimental facilities specifications are shown in Table 1.

3. Decision of Optimum Finishing Conditions

Experiments were performed in three sections: searching for the optimal normal force, finding the optimal tool angle, and polishing a limited area in an iterative process under the optimal conditions[3],[4].

Table 1 Experimental conditions

Workpiece	
Material	Aluminum alloy 5052
Size	120×160×30 *mm*
Polishing Tool	
Grain	Al_2O_3
Grain size	#120, #320, #500
Bond	Synthetic rubber
Concentration	70%
Type	Cup shape disk
Disk diameter	Out:30 *mm*, In:10 *mm*
Shank diameter	3 *mm*
Polishing Conditions	
Rotational speed(ω)	1650 *rpm*
Linear speed (v)	2.7 *mm/sec*
Normal force (F_z)	2 *N*
Lateral distance (d_w)	1 *mm*
Length of polishing area (L)	10 *mm*, 30 *mm*, 80 *mm*
Width of polishing area (W)	30 *mm*, 50 *mm*
Coolant	Vegetable oil
Path	

3.1 Hand-polishing

In this research work, robotic polishing was performed using a hand-polishing system. As the tool and polishing system are generally used for hand-polishing, it was decided to carry out initial polishing experiments by hand before robotic polishing. After considering the obtained results, the robotic polishing conditions were decided upon.

In the hand-polishing process, the worker controls the polishing force in order to obtain a fine surface. Tool movements are generally in both longitudinal and lateral directions without any regulation. Another conditional point in the hand-polishing process is the angle of the tool relative to the workpiece. In this paper, it is referred to as "Tool Angle" and is represented by "α". During the polishing process, the tool angle is changed slightly by worker, and as such, it is difficult to clearly determine the magnitude of tool angle. Figure 3 shows a change of normal force during the hand-polishing process when the polishing force is completely controlled. As can be seen from this figure, 2*N* is a suitable normal force. Moreover, the continuance of polishing by hand is determined by visually checking the obtained surface brightness.

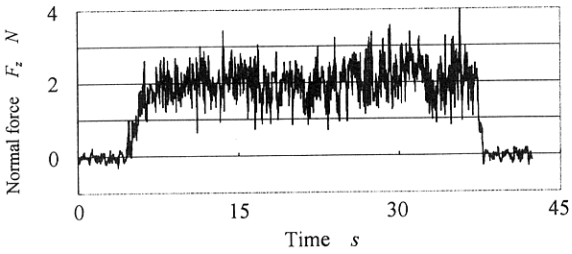


Fig. 3 Normal force during hand-polishing process

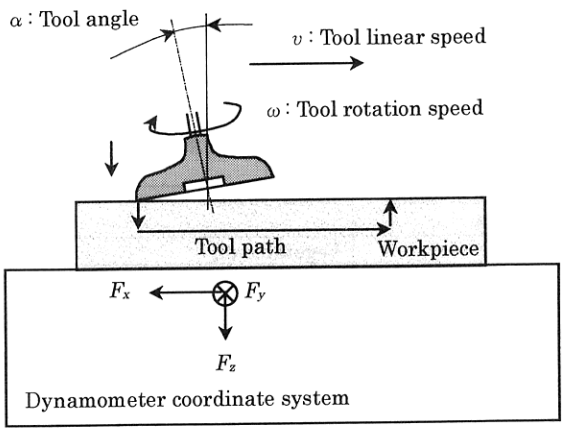


Fig. 4 Method of experiments

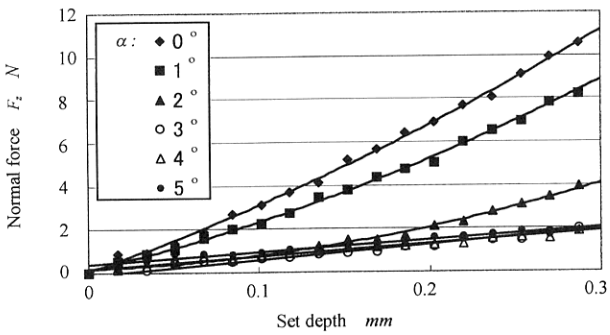


Fig. 5 Relation between set depth and normal force considering the tool angle

3.2 Optimal tool angle

A skilled worker is able to develop a feel for the optimal tool angle through experience. However, measuring that tool angle is not simple; as such, it is first necessary to find out the optimal tool angle. For this purpose, some experiments were performed using the robot under the conditions shown in Fig. 4. The tool path was a straight path parallel to the workpiece surface. The normal force at the start point and the end point of path was set the same. Setting the normal force was performed while the tool was stationary. In this series of experiments, the normal force was set to $2N$, as ascertained from hand-polishing experiments. It can be seen from Fig. 5 that this normal force is equivalent to 50 μm～300 μm in set depth with regard to the tool angle. This change in set depth is due to elasticity of the tool and flexibility of the robot hand, although the former is much more effective than the latter. Therefore, as seen in this diagram, given the same normal force, the equivalent set depth is different according to the tool angle. The experiments were performed using various tool angles from 1° to 5° at equal intervals of 1° and two other angles of 10° and 20° were also examined. Figure 6 shows the normal force obtained from a series of these experiments using the robot. As this figure shows, the average normal force is $2N$. The length of traversing was set to 10 mm and this process was carried out 10 times in the same path (lateral distance $d_w=0$). Each step of the experiment was repeated seven times, and the mean of the data obtained in these seven trials was taken into account. Three components of polishing force, depth of machined groove, and surface roughness were the main results obtained from these experiments.

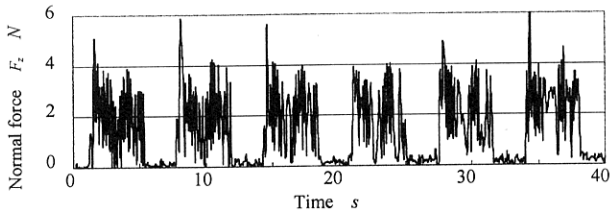


Fig. 6 Normal force in process of polishing by robot ($\alpha=2°$)

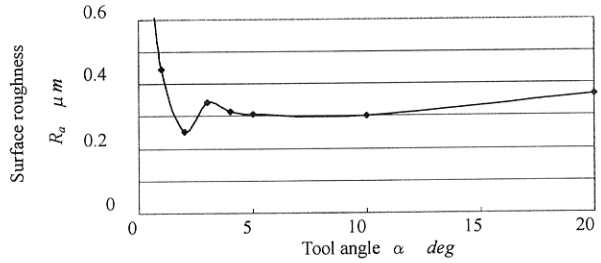


Fig. 7 Relation between tool angle and surface roughness

The most proper parameter to consider when deciding upon tool angle is surface roughness. As a result, the relation between tool angle and surface roughness was considered to be a good criterion for the tool angle decision. Figure 7 shows this relationship. As seen in this figure, the minimum surface roughness has been obtained in tool angle $\alpha=2°$. Figure 8 shows the relation between tool angle and groove depth and width. An increase in the tool angle causes an increase in depth and a decrease in width. For the most efficient surface polishing process, less depth and more width are desired. It means that when the same surface roughness is obtained from two or more tool angles, then the least tool angle is the best. Based on the results, $\alpha=2°$ is the optimal tool angle;

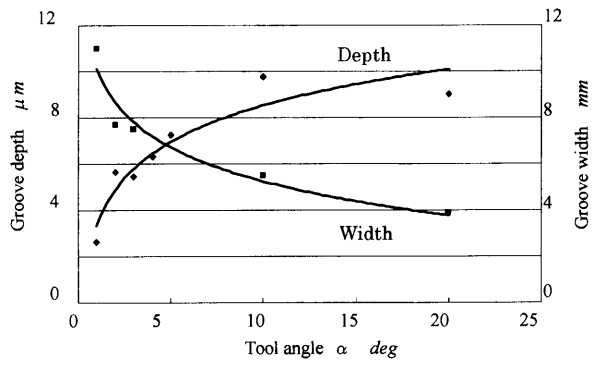


Fig. 8 Relation between tool angle and groove depth and width

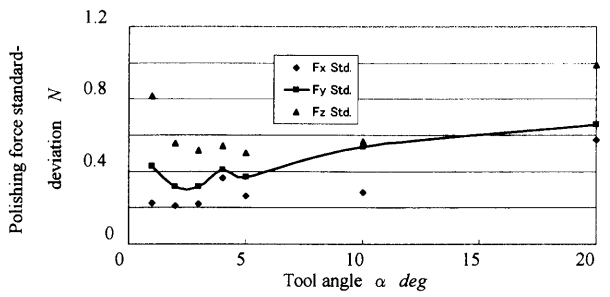


Fig. 9 Relation between tool angle and polishing force standard deviation

however, considering the inclination of disk surface β, the real angle between disk surface and workpiece is $\alpha-\beta$ that is about 1°.

In order to build an autonomous agent for the robot to grind a flat surface, it is necessary to use proper feedback data from the environment to the agent. In this series of experiments, we were interested in finding a relationship between the polishing force and surface roughness. As can be seen in Fig. 9, the standard deviation of polishing force is the most proper parameter for this purpose. Among the three components of polishing force, the standard deviation of F_y behaves in a manner most similar to the graph shown in Fig. 7. Besides surface roughness, this parameter of polishing force can also be used as an in-process feedback for making the autonomous agent.

3.3 Number of polishing strokes

Polishing is an iterative process and the number of polishing strokes is an important parameter in obtaining good results. In the case of hand-polishing, the surface roughness is estimated during the process by a skilled worker visually. In this way the worker can decide the optimal number of polishing strokes. In the robotic polishing process, a procedure for optimal number of polishing strokes is also necessary. Therefore, in order to find the optimal number of polishing strokes, using the proper tool angle obtained from previous series of experiments, a series of iterative surface polishing tests were performed. In this series of experiments, the relationship between surface roughness R_a and number of polishing strokes N_P was investigated. We used three different grain size polishing disks with grain sizes of #120, #320 and #500, the same as used by a skilled worker. The process was repeated 200 times for each of the tools while the normal force was set to $2N$.

Figure 10 shows the relation between the number of polishing strokes N_P and surface roughness R_a for the disks #120, #320 and #500. In this figure, solid lines and broken lines are two kinds of data approximation that will be explained in next section. Disk #120 was used for initial polishing on initial surface that was machined by a milling machine. Disks #320 and #500 were used for polishing on surfaces with surface roughness under 0.1 μm because they are used for surface polishing following the primary polishing using disk #120. As can be seen in Fig. 10, surface roughness begins to decrease and after $N_P=50$ strokes results in an improvement of roughness. However, continuation of the process, does not considerably improve surface roughness. In other words, repeating the process more than 50 times will only cause an increase in groove depth. Moreover, because of the tool being worn down after a number of polishing strokes, tool life is consumed gradually. This has a direct correlation to increased surface roughness. The increasing of surface roughness at final polishing steps using tool #320 and #500 in Fig. 10 may have occurred as a result of this phenomenon.

On the other hand, the relation between number of polishing strokes and groove depth is shown in Fig. 11. As can be seen from this figure, the relationship between number of polishing strokes N_P and groove depth d_g can be described using two straight lines with different slopes. The intersection point of these lines occurred at the number of polishing strokes that is named critical number of polishing strokes ($N_{Pc}=50$). In this point, about 90%~99% of final improved surface can be obtained. As a result, the best surface quality is obtained by repeating the process around 50 times under the conditions presented in Table 1.

4. Multi-stage Polishing

As explained in the previous section, the surface roughness R_a has almost been improved at N_{Pc} strokes of polishing that is the number of polishing strokes at the intersection point of depth diagram in Fig. 11. This situation is similar to the results of Sasaki et al.[5], that were carried out using whetstone. In that report, the relation between R_a and N_P was expressed using an exponential function. Therefore, we tried to

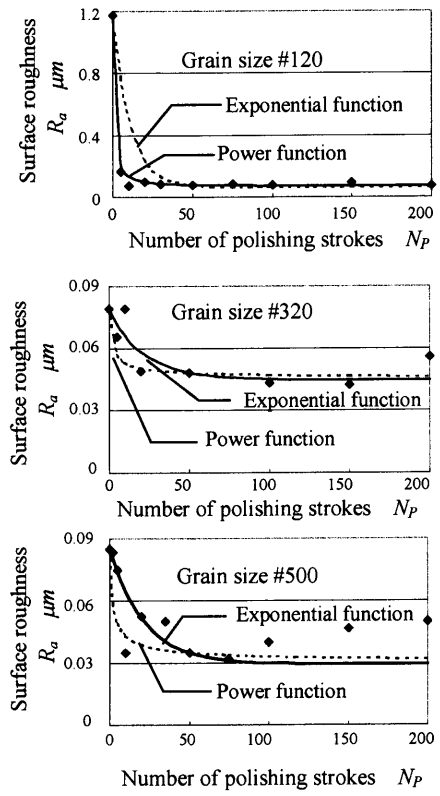


Fig. 10 Relation between the number of polishing strokes and surface roughness

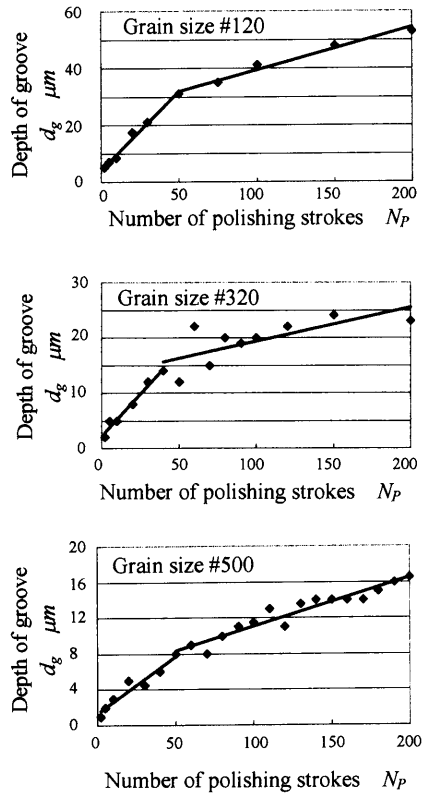


Fig. 11 Relation between the number of polishing strokes and groove depth

propose a process model for expressing the relation between R_a and N_P. However, in this study we examined two models of exponential function and power function for expressing this relationship. In order to estimate the effectiveness of the applied models in practice, calculated results based on the models were then compared with experimental results.

In the case of the power function, the relation between R_a and N_P can be described using the following equation:

$$R_a=(R_0-R_e)(N_P+1)^{-b_P}+R_e \quad (1)$$

Where R_0 is the initial surface roughness and R_e is the final surface roughness. In this equation, the parameters b_P and R_e are two unknown parameters. For determining them, notice that when $N_P=N_{Pc}$ the surface roughness is improved around 90%~99%. Therefore,

$$(N_{Pc}+1)^{-b_P}=0.01\sim0.1 \quad (2)$$

From Eq.(2), b_P can be derived. Then, from Eq.(1), by substituting the value of R_a at $N_{Pc}=50$, R_e can be obtained.

In the same way, the relation between R_a and N_P in the case of the exponential function can be described using the following equation:

$$R_a=(R_0-R_e)e^{-b_eN_P}+R_e \quad (3)$$

When $N_P=N_{Pc}$ the surface roughness is improved around 90%~99%. Therefore,

$$e^{-b_eN_{Pc}}=0.01\sim0.1 \quad (4)$$

and then b_e and R_e can be derived from Eqs.(4) and (3).

Using the above process models, calculated results of the R_a and N_P relationship are shown in Fig. 10. The suitable applied model for each polishing disk is shown with a solid line, and the other model is represented by a broken line. As can be seen from this figure, a power function is a suitable model for polishing using polishing disk #120 on an initial machined surface by milling machine. However it is not used for polishing disks #320 and #500. For these cases, an exponential function is more suitable.

Moreover, a practical mirror polishing was performed using the robot and calculated results were compared with experimental results. In this practical process, an area of 50 mm×80 mm was polished in three iterative polishing stages on a workpiece with initial surface roughness of $R_a=1.2$ μm. In this multi-stage polishing process, three different grain size of polishing disks of #120, #320 and #500 were used in that order. The number of polishing strokes selected for each stage was set at 50, based on an obtained N_{Pc} value as explained in the previous section. A separate process model was applied for each stage in order to estimate the relation between surface roughness and

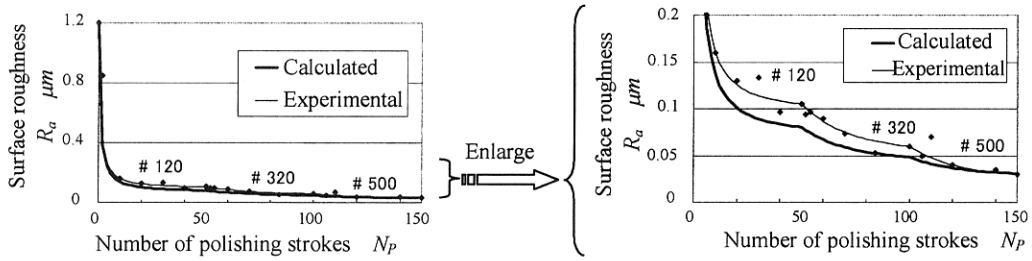


Fig. 12 Comparison between calculated and experimental results in multi-stage polishing

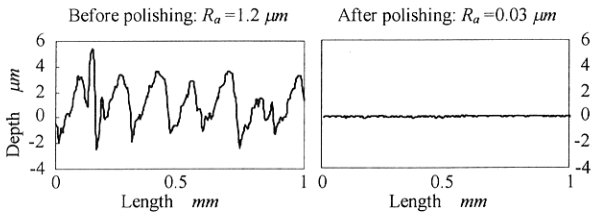


Fig. 13 Comparison of surface roughness before and after the process

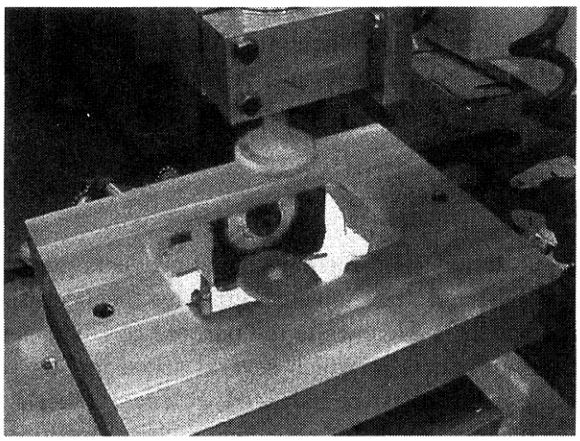


Fig. 14 Finally obtained mirror-polished surface

number of strokes. Figure 12 shows both calculated results and experimental results. As this figure shows, the calculated values were confirmed by experimental results.

Figure 13 shows the comparison between surface roughness of workpiece before and after the process, and Fig. 14 shows this mirror-polished area on the workpiece. It has been confirmed that good results can be obtained using this method.

5. Conclusions

The application of a six-axis articulated robot for polishing an aluminum alloy at constant pressure has been studied and the relevant data are presented. The polishing has been performed by polishing disks of three grain sizes. From the experiments, the obtained results and optimal machining conditions and the integrity of mirror surface are summarized in the following:

1. Robotic polishing was carried out using the same normal force as hand-polishing for some various tool angles. From those experiments, suitable tool angle was obtained.

2. As the number of polishing strokes was increased, the roughness of the polished surface became improved. The obtained data was approximated using an exponential equation and a power equation. The surface roughness R_a approaches final surface roughness R_e after the critical number of polishing strokes N_{Pc}.

3. For practical mirror polishing, a multi-stage polishing process was carried out. It was confirmed that a good mirror-polished surface can be produced using the proposed robotic polishing system method by selecting the optimal machining parameters obtained in the experiments. In addition, the effectiveness of the applied models was confirmed by experimental results during the multi-stage polishing process.

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